Introducing virtual work

Actual work:

or $\int \vec{F} \cdot d\vec{r}$

$$W = \sum_i f_i \cdot dr_i$$

f_i: actual force; dr_i: actual displacement

$$r_i = r_i(q, q_1, q_j)$$

$$\delta r_i^* = \sum_j \frac{\partial r_i^*}{\partial q_j} \delta q_j$$

Virtual work:

$$\delta W^* = \sum_i f_i^* \cdot \delta r_i^*$$

f_i^*: virtual force

$$\delta W^* = \sum_i f_i^* \cdot \delta r_i^* = \sum_i f_i^* \cdot \sum_j \left(\frac{\partial r_i^*}{\partial q_j} \delta q_j \right)$$

$$= \sum_j \left(\sum_i f_i \cdot \frac{\partial r_i^*}{\partial q_j} \right) \delta q_j$$

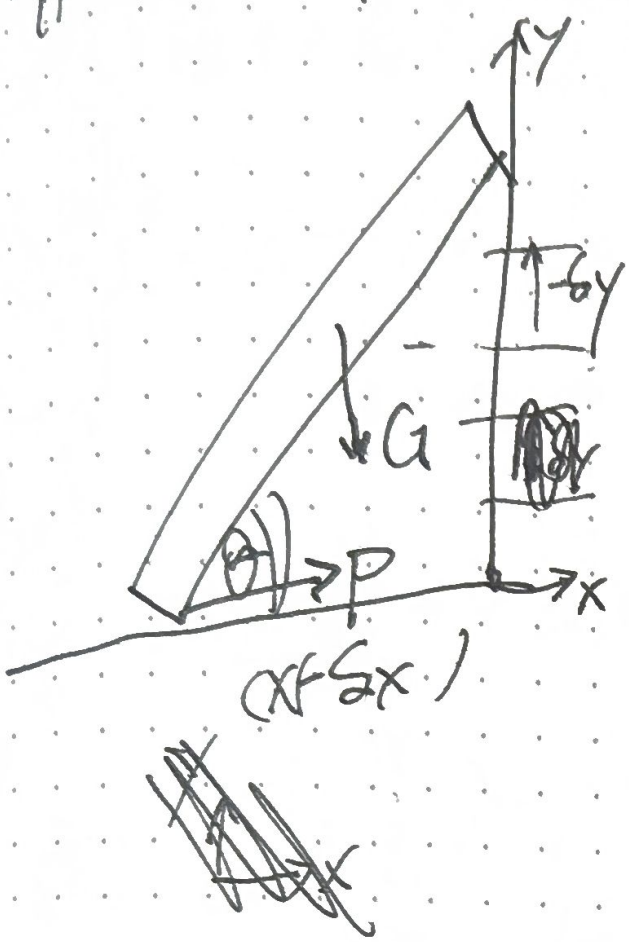
Q_j

def: $Q_j = \sum_i f_i \cdot \frac{\partial r_i^*}{\partial q_j}$

$$\delta W^* = \sum_j Q_j \delta q_j$$

Q_j is kind a like the conjugated force through conjugate $\dot{p}_j$ from Hamiltonian.

Application of Virtual work.



If equarbbmm

$\delta W = 0$

$\delta W = P \cdot \delta r_P + G \cdot \delta r_G$

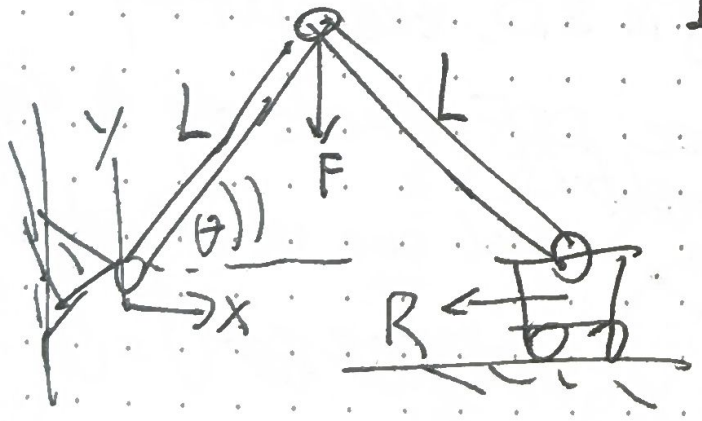
$= P\delta X + G(-\delta y) = 0$

$P = G\frac{\delta y}{\delta W}$

$x = \frac{L}{2}\cos\theta \Rightarrow \delta x = L\sin\theta\delta\theta$

$y = \frac{L}{2}\sin\theta \quad \delta y = \frac{L}{2}\cos\theta\delta\theta$

$P = G\frac{\frac{L}{2}\cos\theta\delta\theta}{L\sin\theta\delta\theta} = \frac{G\cos\theta}{2\sin\theta}$



If equilibrium

$\delta W = 0$

$F\delta y - R\delta X = 0$

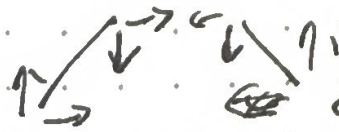
$y = l\sin\theta \quad \delta y = l\cos\theta\delta\theta$

$X = 2l\cos\theta \quad \delta X = -2l\sin\theta\delta\theta$

$Fl\cos\theta\delta\theta + 2Rl\sin\theta\delta\theta = 0$

$F = -2\frac{\sin\theta}{\cos\theta}R.$

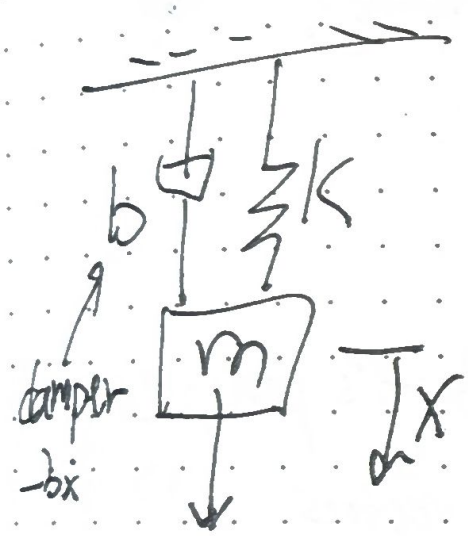
If solving using ~~motor~~ newton method.



Let's at time, ~~Let's at torque~~.

The actual derivation is from D'Alembert's principle, formalized by Lagrange, result in E-L equation. Euler was very impressed!

Example of this generalized force in L mech:



$$T = \frac{1}{2}m\dot{x}^2$$

$$U = \frac{1}{2}kx^2 - mgx$$

$$\sum F_i^{N.C} = F\hat{i} - b\dot{x}\hat{i}$$

$$\mathcal{L} = T - U$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\ddot{x}$$

$$\frac{\partial \mathcal{L}}{\partial x} = kx - mg$$

E-L equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = m\ddot{x} - kx + mg = \underline{Q_x} \; ?$$

$$Q_x = \sum F_i \cdot \delta x$$

$$(F\hat{i} - b\dot{x}\hat{i}) \cdot \delta x = Q_x \delta x = \delta W_x$$ ← N.C. (non conservative) in coordinate of x.

$$Q_x = F(t) - b\dot{x}$$

$$m\ddot{x} - kx + mg = F(t) - b\dot{x}$$

How Lagrange did it when he was 19:

Newton's 2nd Law:

$$\vec{F} = m\vec{a}$$

$$\downarrow$$

$$F_x = ma_x = m\ddot{x} = m(\dot{x}) = (m\dot{x})\dot{} = \dot{p}_x$$

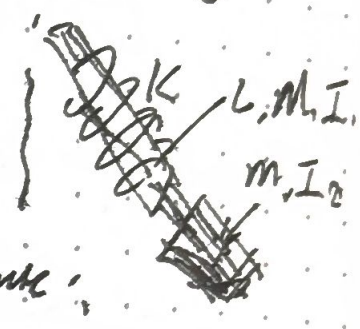
$$F_x = \dot{p}_x$$

$$F_x \delta x = \dot{p}_x \delta x$$

that's one component / coordinate;

if we generalize this for all coordinates:





$$\sum_i F_i \cdot \delta r_i = \sum_i p_i \cdot \delta r_i$$

$$\downarrow$$

$$\sum_i (F_i - p_i) \cdot \delta r_i = 0 \leftarrow \text{D'Alembert's principle}$$

dot product to decompose vectors

$$\delta W = \sum_i F_i \cdot \delta r_i = \sum_j Q_j \delta q_j = \sum_i \dot{p}_i \cdot \delta r_i$$

To simplify the system, let's look at one particle on one coordinate, $x = x(q,t)$

$$\sum Q_1 \delta q_1 = \sum \dot{p} \delta x = \sum m\ddot{x} \delta x = m\ddot{x} \frac{\partial x}{\partial q} \delta q = m \delta q \ddot{x} \frac{\partial x}{\partial q} \quad (1)$$

envoke calc magic #1: production rule

$$d(fg) = g\,df + f\,dg$$

$$d(\dot{x}x') = x'\ddot{x} + \dot{x}\frac{dx'}{dt}$$

$$\frac{d}{dt}\left(\dot{x}\frac{\partial x}{\partial q}\right) = \frac{\partial x}{\partial q}\ddot{x} + \dot{x}\frac{d}{dt}\left(\frac{\partial x}{\partial q}\right) \rightarrow \ddot{x}\frac{\partial x}{\partial q} = \frac{d}{dt}\left(\dot{x}\frac{\partial x}{\partial q}\right) - \dot{x}\frac{d}{dt}\left(\frac{\partial x}{\partial q}\right)$$

(1) → $m\frac{\partial x}{\partial q}\left(\frac{d}{dt}\left(\dot{x}\frac{\partial x}{\partial q}\right) - \dot{x}\frac{d}{dt}\left(\frac{\partial x}{\partial q}\right)\right)$

$= \frac{\partial}{\partial q}\left[\sum\left(m\frac{d}{dt}\left(\dot{x}\frac{\partial x}{\partial q}\right) - m\dot{x}\frac{d}{dt}\left(\frac{\partial x}{\partial q}\right)\right)\right]$

(2) (3)

+Σ if there's more than one coordinate still yields $\vec{0}$ - $\vec{0}$ at every coordinate

envoke calc magic #2: $\frac{\partial \dot{x}}{\partial \dot{q}} = \frac{\partial x}{\partial q}$ ←

$\dot{x} = \frac{\partial x}{\partial q}\dot{q} + \frac{\partial x}{\partial t}$

$\frac{\partial \dot{x}}{\partial \dot{q}} = \frac{\partial x}{\partial q} + \frac{\partial}{\partial \dot{q}}\left(\frac{\partial x}{\partial t}\right)$

(2) → $m\frac{d}{dt}\left(\dot{x}\frac{\partial x}{\partial q}\right) = m\frac{d}{dt}\left(\dot{x}\frac{\partial \dot{x}}{\partial \dot{q}}\right) = \frac{d}{dt}\left(m\dot{x}\frac{\partial \dot{x}}{\partial q}\right)$

$= \frac{d}{dt}\left(\frac{\partial}{\partial \dot{q}}\left(\frac{1}{2}m\dot{x}^2\right)\right)$ def $T = \frac{1}{2}m\dot{x}^2$

$= \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}}\right)$

envoke calc magic #3: $\frac{d}{dt}\frac{\partial x}{\partial q} = \frac{\partial}{\partial q}\frac{dx}{dt} = \frac{\partial \dot{x}}{\partial q}$

multi-variable derivative:

$\frac{d}{dt}f(q,t) = \frac{\partial f}{\partial q}\frac{dq}{dt} + \frac{\partial f}{\partial t}$ $\frac{d}{dt}(x) = \frac{\partial x}{\partial q}\frac{dq}{dt} + \frac{\partial x}{\partial t} = \frac{\partial x}{\partial q}\dot{q} + \frac{\partial x}{\partial t}$

$\frac{d}{dt}\left(\frac{\partial x(q,t)}{\partial q}\right) = \frac{\partial^2 x}{\partial q^2}\frac{\partial q}{\partial t} + \frac{\partial^2 x}{\partial t \partial q}$

$\frac{\partial^2 x}{\partial q \partial t} = \frac{\partial^2 x}{\partial t \partial q}$ calc magic #4

$= \frac{\partial}{\partial q}\left(\frac{\partial x}{\partial q}\dot{q} + \frac{\partial x}{\partial t}\right) = \frac{\partial \dot{x}}{\partial q}$

($\dot{x}$)

(3) → $m\dot{x}\frac{d}{dt}\left(\frac{\partial x}{\partial q}\right) = m\dot{x}\frac{\partial \dot{x}}{\partial q} = \frac{\partial}{\partial q}\left(\frac{1}{2}m\dot{x}^2\right) = \frac{\partial}{\partial q}(T)$

$$Q^{NC}\cdot\delta q + Q^{C}\delta q = \left[\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}}\right) - \frac{\partial T}{\partial q}\right]\delta q$$

$$Q^{C} = -\frac{\partial V}{\partial q} \qquad \boxed{\frac{\partial V}{\partial \dot{q}} = 0} \text{ for classical mechanic systems.}$$

$$Q^{NC}\delta q = \left(\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}}\right) - \frac{\partial T}{\partial q} + \frac{\partial V}{\partial q}\right)\delta q$$

$$Q^{NC}\delta q = \left[\frac{d}{dt}\left(\frac{\partial}{\partial \dot{q}}(T-V)\right) - \frac{\partial}{\partial q}(T-V)\right]\delta q$$

def. $\mathcal{L} = T - V$

$$Q^{NC}\delta q = \left(\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q}\right)\delta q$$

$$Q_q^{NC} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q}$$